

# Skill registry — 2026-05-26

A register of every custom skill I've written for Claude Code (Anthropic's coding assistant CLI), and whether it has measured data behind its claims yet. A "skill" here is a reusable slash command — a named recipe Claude Code runs when I type `/audit` or similar. 4 repos, 35 skills, 34 of them A/B-tested for token savings so far.

## A/B-MEASURED SAVE RATES BY PORTFOLIO

**+23%** +78K tokens across 8 skills

**AudiobookMaker** (arm A 342K → arm B 264K)



**+21%** +27K tokens across 4 skills

**mikkonumminen.dev** (arm A 128K → arm B 101K)



**+20%** +170K tokens across 11 skills

**Spacepotatis** (arm A 853K → arm B 683K)



**+7%** +40K tokens across 11 skills

**claude-skills** (arm A 590K → arm B 550K)



**Aggregate: +17% (+316K tokens across 34 skills).** Save rate = (arm A – arm B) / arm A, summed per portfolio. Negative means the skill costs MORE per use than going cold — those skills encode rigor (audit thoroughness, protocol discipline, spec depth), not scout-savings. The 3× heuristic baseline assumed everything would save ~67%; measurement says the truth varies from +52% to –112% depending on what the skill encodes.

Per-skill, four numbers per row: **measured cost / use** (real tokens billed when a skill ran), **measured save / use** (A/B test: arm A minus arm B), **estimated cost / use** (the author's pre-measurement guess), and **estimated save / use** (the 3× heuristic applied to the estimated cost). Green = measured. Italic gray = estimated. Orange measured-save means the skill cost MORE in the A/B than the unstructured baseline — real findings, not bugs. Cost-or-save dashes (—) mean no data exists for that cell yet.

Per-repo summary

| REPO              | SKILLS | COST / USE MEASURED | SAVE / USE MEASURED | ESTIMATE-ONLY |
|-------------------|--------|---------------------|---------------------|---------------|
| AudiobookMaker    | 8      | 8                   | 8                   | 0             |
| Spacepotatis      | 12     | 11                  | 11                  | 1             |
| claude-skills     | 11     | 11                  | 11                  | 0             |
| mikkonumminen.dev | 4      | 4                   | 4                   | 0             |

All measurement counts are per single invocation of the skill. Cost-measured rows have a real run behind their per-use cost (a transcript-attributed session, or an A/B calibration arm-B). Save-measured rows have a real A/B test behind their per-use savings (calibration). Estimate-only rows have neither yet.

Claude Code built-ins

reference — not part of the portfolio

Built-in slash commands. Per-use cost is measured from real session transcripts the same way as the custom-skill rows. Per-use save is measured by treating the built-in's prompt as an arm-B recipe in the same A/B methodology used on custom skills — both measured columns are real numbers, not heuristics. The estimated-cost cell stays "—" because built-ins have no SKILL.md author who wrote a guess; the estimated-save cell shows the project-wide 3× baseline heuristic prediction (2× measured cost) so the row carries the same heuristic-vs-measured contrast the custom-skill rows show. Methodology and raw arm-A / arm-B numbers: docs/audits/builtin-review-calibration-2026-05-22.md .

| SKILL                                          | STATUS                       | COST / USE<br><i>measured</i> | SAVE / USE<br><i>measured</i> | COST / USE<br><i>estimated</i> | SAVE / USE<br><i>estimated</i>       |
|------------------------------------------------|------------------------------|-------------------------------|-------------------------------|--------------------------------|--------------------------------------|
| /review<br>Claude Code built-in PR code review | ✓MEASURED<br>last 2026-05-21 | 1.05M<br>tokens / use         | 46K<br>tokens / use (63%)     | —                              | 2.10M<br>tokens / use (3× heuristic) |

## Calibration honesty — where my guesses landed

Here are the rows where I had a guess before I had data. The “How wrong” column is the measurement divided by the guess. Green means I landed within  $\pm 10\%$ . Orange means I was off by 5 $\times$  or more in either direction. 17 of 32 rows are orange. The fix is not to write better guesses next time. The fix is to keep measuring.

| SKILL                                      | MY GUESS (PER USE) | MEASURED (PER USE) | HOW WRONG          |
|--------------------------------------------|--------------------|--------------------|--------------------|
| AUDIOBOOKMAKER<br>release-cut              | 2K                 | 188K               | 111 $\times$ under |
| SPACEPOTATIS<br>equipment                  | 4K                 | 401K               | 93 $\times$ under  |
| CLAUDE-SKILLS<br>mikko-help                | 7K                 | 644K               | 92 $\times$ under  |
| CLAUDE-SKILLS<br>session-cost              | 8K                 | 409K               | 51 $\times$ under  |
| SPACEPOTATIS<br>modular-architecture-audit | 5K                 | 129K               | 26 $\times$ under  |
| AUDIOBOOKMAKER<br>release-bundle-audit     | 4K                 | 87K                | 23 $\times$ under  |
| AUDIOBOOKMAKER<br>worktree-launch          | 1K                 | 23K                | 18 $\times$ under  |
| AUDIOBOOKMAKER<br>work-session             | 2K                 | 23K                | 14 $\times$ under  |
| AUDIOBOOKMAKER<br>ci-failure-triage        | 2K                 | 25K                | 13 $\times$ under  |
| SPACEPOTATIS<br>new-enemy                  | 6K                 | 68K                | 12 $\times$ under  |
| SPACEPOTATIS<br>new-story                  | 5K                 | 56K                | 10 $\times$ under  |

| SKILL                                             | MY GUESS (PER USE) | MEASURED (PER USE) | HOW WRONG         |
|---------------------------------------------------|--------------------|--------------------|-------------------|
| AUDIOBOOKMAKER<br><b>pronunciation-corpus-add</b> | 2K                 | 17K                | <b>9.5×</b> under |
| CLAUDE-SKILLS<br><b>mikko-install</b>             | 4K                 | 35K                | <b>8.8×</b> under |
| SPACEPOTATIS<br><b>new-mission</b>                | 8K                 | 50K                | <b>6.2×</b> under |
| CLAUDE-SKILLS<br><b>skill-usage</b>               | 4K                 | 25K                | <b>6.2×</b> under |
| SPACEPOTATIS<br><b>new-perk</b>                   | 9K                 | 56K                | <b>6.2×</b> under |
| CLAUDE-SKILLS<br><b>skill-calibration</b>         | 140K               | 27K                | <b>5.1×</b> over  |
| SPACEPOTATIS<br><b>new-solar-system</b>           | 13K                | 60K                | <b>4.6×</b> under |
| AUDIOBOOKMAKER<br><b>copyright-scan</b>           | 3K                 | 14K                | <b>4.5×</b> under |
| AUDIOBOOKMAKER<br><b>voice-pack-finnish</b>       | 6K                 | 27K                | <b>4.3×</b> under |
| MIKKONUMMINEN.DEV<br><b>skill-localUpdate</b>     | 83K                | 21K                | <b>3.9×</b> over  |
| SPACEPOTATIS<br><b>content-audit</b>              | 15K                | 53K                | <b>3.5×</b> under |
| SPACEPOTATIS<br><b>new-migration</b>              | 7K                 | 21K                | <b>2.9×</b> under |
| CLAUDE-SKILLS<br><b>ai-codegen-smell-audit</b>    | 75K                | 199K               | <b>2.7×</b> under |
| SPACEPOTATIS<br><b>balance-review</b>             | 14K                | 33K                | <b>2.5×</b> under |

| SKILL                                             | MY GUESS (PER USE) | MEASURED (PER USE) | HOW WRONG                     |
|---------------------------------------------------|--------------------|--------------------|-------------------------------|
| CLAUDE-SKILLS<br><b>readme-drift-sync</b>         | 40K                | 63K                | <b>1.6×</b> under             |
| SPACEPOTATIS<br><b>save-roundtrip-audit</b>       | 12K                | 19K                | <b>1.5×</b> under             |
| CLAUDE-SKILLS<br><b>audit</b>                     | 425K               | 553K               | <b>1.3×</b> under             |
| MIKKONUMMINEN.DEV<br><b>md-to-pdf</b>             | 25K                | 21K                | <b>1.2×</b> over              |
| MIKKONUMMINEN.DEV<br><b>sync-readmes</b>          | 141K               | 119K               | <b>1.2×</b> over              |
| MIKKONUMMINEN.DEV<br><b>skill-registry</b>        | 130K               | 148K               | <b>1.1×</b> under             |
| CLAUDE-SKILLS<br><b>react-anti-patterns-audit</b> | 23K                | 22K                | <b>1.0×</b> within $\pm 10\%$ |

## Multi-model calibration — does the save rate hold across models?

Same A/B methodology as the per-repo Sonnet calibrations, repeated with sub-agents dispatched as Opus and (later) Haiku. Each skill below was measured at least twice — once per model. The save rate is what changes: Opus is more efficient cold, so the recipe's value compresses; Sonnet rewards scaffolding skills more. Sorted by spread across models — biggest model-sensitivity first.

| SKILL                                     | SONNET                                              | OPUS                                                          | HAIKU                                              |
|-------------------------------------------|-----------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------|
| CLAUDE-SKILLS<br><b>readme-drift-sync</b> | <b>−112%</b><br>arm A 30K → B 63K<br>−33K saved/use | <b>−4%</b><br>arm A 54K → B 56K<br>−2K saved/use              | <b>−6%</b><br>arm A 41K → B 44K<br>−3K saved/use   |
| CLAUDE-SKILLS<br><b>session-cost</b>      | <b>+14%</b><br>arm A 36K → B 31K<br>5K saved/use    | <b>+4%</b><br>arm A 33K → B 32K<br>1K saved/use               | <b>−89%</b><br>arm A 30K → B 56K<br>−26K saved/use |
| CLAUDE-SKILLS<br><b>skill-calibration</b> | <b>−74%</b><br>arm A 16K → B 27K<br>−12K saved/use  | <b>+13%</b><br>arm A 42K → B 36K<br>6K saved/use              | <b>+15%</b><br>arm A 38K → B 32K<br>6K saved/use   |
| CLAUDE-SKILLS<br><b>skill-usage</b>       | <b>+53%</b><br>arm A 49K → B 23K<br>26K saved/use   | <b>−14%<sup>†</sup></b><br>arm A 32K → B 37K<br>−5K saved/use | <b>+57%</b><br>arm A 64K → B 28K<br>36K saved/use  |
| SPACEPOTATIS<br><b>equipment</b>          | <b>−5%</b><br>arm A 89K → B 94K<br>−5K saved/use    | <b>+65%</b><br>arm A 95K → B 33K<br>61K saved/use             | <b>−3%</b><br>arm A 44K → B 45K<br>−1K saved/use   |
| SPACEPOTATIS<br><b>new-solar-system</b>   | <b>+9%</b><br>arm A 65K → B 60K<br>6K saved/use     | <b>+71%</b><br>arm A 97K → B 28K<br>70K saved/use             | <b>+16%</b><br>arm A 48K → B 41K<br>8K saved/use   |
| SPACEPOTATIS<br><b>new-migration</b>      | <b>+16%</b><br>arm A 25K → B 21K<br>4K saved/use    | <b>+25%</b><br>arm A 43K → B 32K<br>11K saved/use             | <b>−33%</b><br>arm A 33K → B 44K<br>−11K saved/use |

| SKILL                                             | SONNET                                              | OPUS                                               | HAIKU                                              |
|---------------------------------------------------|-----------------------------------------------------|----------------------------------------------------|----------------------------------------------------|
| SPACEPOTATIS<br><b>new-enemy</b>                  | <b>+9%</b><br>arm A 75K → B 68K<br>7K saved/use     | <b>+65%</b><br>arm A 79K → B 28K<br>51K saved/use  | <b>+57%</b><br>arm A 62K → B 27K<br>35K saved/use  |
| SPACEPOTATIS<br><b>modular-architecture-audit</b> | <b>+26%</b><br>arm A 175K → B 129K<br>46K saved/use | <b>-21%</b><br>arm A 69K → B 83K<br>-14K saved/use | <b>+29%</b><br>arm A 46K → B 33K<br>13K saved/use  |
| CLAUDE-SKILLS<br><b>ai-codegen-smell-audit</b>    | <b>+48%</b><br>arm A 157K → B 81K<br>76K saved/use  | <b>+47%</b><br>arm A 172K → B 92K<br>81K saved/use | <b>+1%</b><br>arm A 61K → B 60K<br>496 saved/use   |
| AUDIOBOOKMAKER<br><b>copyright-scan</b>           | <b>+49%</b><br>arm A 65K → B 33K<br>32K saved/use   | <b>+16%</b><br>arm A 48K → B 40K<br>8K saved/use   | <b>+57%</b><br>arm A 61K → B 27K<br>35K saved/use  |
| CLAUDE-SKILLS<br><b>mikko-help</b>                | <b>-22%</b><br>arm A 24K → B 29K<br>-5K saved/use   | <b>-21%</b><br>arm A 33K → B 40K<br>-7K saved/use  | <b>-62%</b><br>arm A 20K → B 33K<br>-12K saved/use |
| MIKKONUMMINEN.DEV<br><b>sync-readmes</b>          | <b>-11%</b><br>arm A 33K → B 37K<br>-4K saved/use   | <b>+17%</b><br>arm A 58K → B 48K<br>10K saved/use  | <b>-24%</b><br>arm A 46K → B 56K<br>-11K saved/use |
| AUDIOBOOKMAKER<br><b>release-bundle-audit</b>     | <b>-12%</b><br>arm A 78K → B 87K<br>-9K saved/use   | <b>+25%</b><br>arm A 94K → B 70K<br>24K saved/use  | <b>+20%</b><br>arm A 52K → B 41K<br>11K saved/use  |
| SPACEPOTATIS<br><b>balance-review</b>             | <b>+28%</b><br>arm A 46K → B 33K<br>13K saved/use   | <b>+40%</b><br>arm A 46K → B 28K<br>19K saved/use  | <b>+5%</b><br>arm A 38K → B 36K<br>2K saved/use    |
| AUDIOBOOKMAKER<br><b>release-cut</b>              | <b>+17%</b><br>arm A 35K → B 29K<br>6K saved/use    | <b>+51%</b><br>arm A 67K → B 33K<br>34K saved/use  | <b>+29%</b><br>arm A 55K → B 39K<br>16K saved/use  |

| SKILL                                             | SONNET                                            | OPUS                                               | HAIKU                                             |
|---------------------------------------------------|---------------------------------------------------|----------------------------------------------------|---------------------------------------------------|
| AUDIOBOOKMAKER<br><b>ci-failure-triage</b>        | <b>+43%</b><br>arm A 44K → B 25K<br>19K saved/use | <b>+10%</b><br>arm A 31K → B 28K<br>3K saved/use   | <b>+14%</b><br>arm A 44K → B 38K<br>6K saved/use  |
| CLAUDE-SKILLS<br><b>mikko-install</b>             | <b>−35%</b><br>arm A 17K → B 23K<br>−6K saved/use | <b>−2%</b><br>arm A 27K → B 27K<br>−503 saved/use  | <b>−14%</b><br>arm A 23K → B 26K<br>−3K saved/use |
| AUDIOBOOKMAKER<br><b>pronunciation-corpus-add</b> | <b>+12%</b><br>arm A 19K → B 17K<br>2K saved/use  | <b>+39%</b><br>arm A 42K → B 25K<br>17K saved/use  | <b>+7%</b><br>arm A 24K → B 22K<br>2K saved/use   |
| AUDIOBOOKMAKER<br><b>voice-pack-finnish</b>       | <b>+52%</b><br>arm A 56K → B 27K<br>29K saved/use | <b>+20%</b><br>arm A 45K → B 36K<br>9K saved/use   | <b>+32%</b><br>arm A 48K → B 33K<br>15K saved/use |
| MIKKONUMMINEN.DEV<br><b>md-to-pdf</b>             | <b>+41%</b><br>arm A 35K → B 21K<br>15K saved/use | <b>+22%</b><br>arm A 36K → B 28K<br>8K saved/use   | <b>+54%</b><br>arm A 59K → B 27K<br>32K saved/use |
| SPACEPOTATIS<br><b>save-roundtrip-audit</b>       | <b>+10%</b><br>arm A 70K → B 63K<br>7K saved/use  | <b>+40%</b><br>arm A 127K → B 77K<br>50K saved/use | <b>+16%</b><br>arm A 70K → B 59K<br>11K saved/use |
| CLAUDE-SKILLS<br><b>mikko-audit-suite</b>         | <b>−21%</b><br>arm A 20K → B 24K<br>−4K saved/use | <b>+5%</b><br>arm A 35K → B 33K<br>2K saved/use    | <b>+8%</b><br>arm A 37K → B 34K<br>3K saved/use   |
| SPACEPOTATIS<br><b>new-mission</b>                | <b>+37%</b><br>arm A 79K → B 50K<br>29K saved/use | <b>+46%</b><br>arm A 79K → B 43K<br>37K saved/use  | <b>+19%</b><br>arm A 41K → B 33K<br>8K saved/use  |
| SPACEPOTATIS<br><b>content-audit</b>              | <b>+22%</b><br>arm A 99K → B 77K<br>22K saved/use | <b>+6%</b><br>arm A 108K → B 102K<br>6K saved/use  | <b>−2%</b><br>arm A 60K → B 61K<br>−912 saved/use |



| SKILL                                             | SONNET                                             | OPUS                                                          | HAIKU                                             |
|---------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------|
| CLAUDE CODE BUILT-IN<br><b>/review</b>            | <b>+63%</b><br>arm A 72K → B 26K<br>46K saved/use  | <b>+39%</b><br>arm A 59K → B 35K<br>23K saved/use             | <b>+52%</b><br>arm A 58K → B 28K<br>30K saved/use |
| AUDIOBOOKMAKER<br><b>worktree-launch</b>          | <b>+1%</b><br>arm A 23K → B 23K<br>126 saved/use   | <b>+24%</b><br>arm A 34K → B 26K<br>8K saved/use              | <b>+23%</b><br>arm A 38K → B 30K<br>9K saved/use  |
| SPACEPOTATIS<br><b>new-perk</b>                   | <b>+32%</b><br>arm A 82K → B 56K<br>27K saved/use  | <b>+53%</b><br>arm A 75K → B 35K<br>40K saved/use             | <b>+30%</b><br>arm A 54K → B 38K<br>16K saved/use |
| CLAUDE-SKILLS<br><b>audit</b>                     | <b>+2%</b><br>arm A 117K → B 114K<br>3K saved/use  | <b>+2%<sup>†</sup></b><br>arm A 214K → B 209K<br>5K saved/use | <b>+23%</b><br>arm A 92K → B 71K<br>21K saved/use |
| MIKKONUMMINEN.DEV<br><b>skill-localUpdate</b>     | <b>+52%</b><br>arm A 44K → B 21K<br>23K saved/use  | <b>+51%</b><br>arm A 59K → B 29K<br>30K saved/use             | <b>+33%</b><br>arm A 46K → B 31K<br>15K saved/use |
| CLAUDE-SKILLS<br><b>security-audit</b>            | <b>-4%</b><br>arm A 108K → B 112K<br>-4K saved/use | <b>-10%</b><br>arm A 124K → B 136K<br>-12K saved/use          | <b>+3%</b><br>arm A 72K → B 69K<br>2K saved/use   |
| CLAUDE-SKILLS<br><b>react-anti-patterns-audit</b> | <b>-33%</b><br>arm A 17K → B 22K<br>-5K saved/use  | <b>-28%</b><br>arm A 24K → B 31K<br>-7K saved/use             | <b>-21%</b><br>arm A 21K → B 25K<br>-4K saved/use |
| SPACEPOTATIS<br><b>new-story</b>                  | <b>+29%</b><br>arm A 47K → B 33K<br>14K saved/use  | <b>+20%</b><br>arm A 77K → B 61K<br>16K saved/use             | <b>+28%</b><br>arm A 60K → B 43K<br>16K saved/use |
| MIKKONUMMINEN.DEV<br><b>skill-registry</b>        | <b>-38%</b><br>arm A 16K → B 22K<br>-6K saved/use  | <b>-39%</b><br>arm A 24K → B 33K<br>-9K saved/use             | <b>-32%</b><br>arm A 20K → B 27K<br>-6K saved/use |

| SKILL          | SONNET                                                                | OPUS                                                                  | HAIKU                                                                 |
|----------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
| AUDIOBOOKMAKER |                                                                       |                                                                       |                                                                       |
| work-session   | <div>-10%</div> <div>arm A 21K → B 23K</div> <div>-2K saved/use</div> | <div>-10%</div> <div>arm A 30K → B 33K</div> <div>-3K saved/use</div> | <div>-12%</div> <div>arm A 24K → B 27K</div> <div>-3K saved/use</div> |

Each cell shows the A/B save rate (% of arm A tokens), the raw arm-A → arm-B tokens, and the absolute saved-per-use. Orange = the skill cost MORE on this model than the unstructured baseline. Cells marked † had a procedure deviation in arm B — see the footnote block below. Sample size N=1 per (skill, model) pair — trust direction and magnitude, not two-significant-digit precision. The cross-portfolio observation is that recipe value shrinks as model capability rises: skills that save 50%+ on Sonnet typically settle at 20-40% on Opus because the cold arm gets cheaper, not because the recipe arm gets more expensive.

† **Procedure deviation** — these 2 measurements are marked with † because the arm-B sub-agent did NOT actually execute the SKILL.md procedure (e.g. tool unavailable in the sandbox, classifier blocked a script). The cost/save number reflects whatever the sub-agent did INSTEAD, not what the skill would actually cost when run normally. Read these rows as "this measurement is compromised" — not as a finding about the skill.

1. **Opus** · CLAUDE-SKILLS skill-usage

— Opus arm B was blocked by the auto-mode classifier from running scan.mjs and ended up using a stale snapshot — did not actually execute the skill procedure. Compare against the Haiku run (clean arm-B, 57% save) for the un-compromised picture.

2. **Opus** · CLAUDE-SKILLS audit

— Opus arm B was blocked from spawning parallel sub-agents (no Agent tool in this sandbox) and executed all 5 audit scopes sequentially in main thread instead of the prescribed parallel fan-out. The cost number reflects sequential execution, not what the skill would actually cost when run normally. Compare against the Haiku run (clean arm-B, 23% save) for the un-compromised picture.

# Per-repo skill tables

One row per skill. Measured rows have a green wash; estimated rows are white. Four data columns per row, in pairs: **Cost / use** measured + estimated, then **Save / use** measured + estimated. Green numbers come from real runs (transcript or A/B). Italic gray numbers are author estimates. Orange numbers are real A/B measurements showing the skill cost MORE per use than the unstructured baseline. Dashes mean no data exists for that cell yet. Every figure is per single invocation — no annual projections anywhere in this document.

| AudiobookMaker                                                                                                                                                                                                                             |                              |                               |                               |                                |                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------|-------------------------------|--------------------------------|--------------------------------|
| 8 skills · 8 measured · <a href="https://github.com/MikkoNumminen/AudiobookMaker">github.com/MikkoNumminen/AudiobookMaker</a>                                                                                                              |                              |                               |                               |                                |                                |
| SKILL                                                                                                                                                                                                                                      | STATUS                       | COST / USE<br><i>measured</i> | SAVE / USE<br><i>measured</i> | COST / USE<br><i>estimated</i> | SAVE / USE<br><i>estimated</i> |
| <b>ci-failure-triage</b><br>When CI fails on a release tag or PR, walk the failure modes in the right order against a recipe library built from the project's fix(ci) commit history.                                                      | ✓MEASURED<br>last 2026-05-23 | 25K<br>tokens / use           | 19K<br>tokens / use           | 2K<br>tokens / use             | 4K<br>tokens / use             |
| <b>copyright-scan</b><br>Scan a git diff (staged by default, or a named revision range / file set) for accidental third-party copyright leaks before they land on origin.                                                                  | ✓MEASURED<br>last 2026-05-12 | 14K<br>tokens / use           | 32K<br>tokens / use           | 3K<br>tokens / use             | 6K<br>tokens / use             |
| <b>pronunciation-corpus-add</b><br>Append a Finnish pronunciation failure (a word the Chatterbox Grandmom voice mispronounces) to the project's pronunciation corpus at docs...                                                            | ✓MEASURED<br>last 2026-05-23 | 17K<br>tokens / use           | 2K<br>tokens / use            | 2K<br>tokens / use             | 4K<br>tokens / use             |
| <b>release-bundle-audit</b><br>Audit the AudiobookMaker PyInstaller release bundle for unused dependencies, dead-code data files, and accidental ML-stack pollution; then propose spec-only fixes on a `chore/release-bundle-size` branch. | ✓MEASURED<br>last 2026-05-23 | 87K<br>tokens / use           | -9K<br>tokens / use           | 4K<br>tokens / use             | 8K<br>tokens / use             |
| <b>release-cut</b><br>Cut a new AudiobookMaker release (bump APP_VERSION, tag vX.Y.Z, push, verify CI lands SHA-256 in release notes AND sidecar). Use this ski...                                                                         | ✓MEASURED<br>last 2026-05-12 | 188K<br>tokens / use          | 6K<br>tokens / use            | 2K<br>tokens / use             | 3K<br>tokens / use             |

|                                                                                                                                                                                                                                                                                              |                              |                     |                     |                    |                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------|---------------------|--------------------|---------------------|
| <b>voice-pack-finnish</b><br>Build a Finnish voice pack from a source audio file end-to-end â€” chunked analyze with ECAPA diarization, transcript validation per speaker, branching to LoRA training (long sources) or few-shot ref-clip packaging (short sources), and ear-check by synth. | ✓MEASURED<br>last 2026-05-23 | 27K<br>tokens / use | 29K<br>tokens / use | 6K<br>tokens / use | 12K<br>tokens / use |
| <b>work-session</b><br>Start, pause, or finish a TODO.md work session as one of the 4 permanent Claude sessions in AudiobookMaker. Use this whenever the user sa...                                                                                                                          | ✓MEASURED<br>last 2026-05-23 | 23K<br>tokens / use | -2K<br>tokens / use | 2K<br>tokens / use | 3K<br>tokens / use  |
| <b>worktree-launch</b><br>Start a new parallel Claude session safely.                                                                                                                                                                                                                        | ✓MEASURED<br>last 2026-05-23 | 23K<br>tokens / use | 126<br>tokens / use | 1K<br>tokens / use | 3K<br>tokens / use  |

| Spacepotatis12 skills · 11 measured · <a href="https://github.com/MikkoNumminen/Spacepotatis">github.com/MikkoNumminen/Spacepotatis</a>                                                                                                                                                                               |                              |                               |                               |                                |                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------|-------------------------------|--------------------------------|--------------------------------|
| SKILL                                                                                                                                                                                                                                                                                                                 | STATUS                       | COST / USE<br><i>measured</i> | SAVE / USE<br><i>measured</i> | COST / USE<br><i>estimated</i> | SAVE / USE<br><i>estimated</i> |
| <b>balance-review</b><br>Diff uncommitted changes to game data (weapons, enemies, waves, missions, perks, augments, loot pools, solar systems) and report DPS, TTK, energy-cost-per-DPS, augment-folded effective DPS, loot-pool roster shifts, and drop-rate deltas vs HEAD.                                         | ✓MEASURED<br>last 2026-05-22 | 33K<br>tokens / use           | 13K<br>tokens / use           | 14K<br>tokens / use            | 27K<br>tokens / use            |
| <b>content-audit</b><br>Pre-commit content invariants check â€” orphan refs (enemy / weapon / sprite / pod / loot-pool / mission system / story trigger), missing sprite generators, perk drop-weight sanity, mission prereq DAG, story integrity (voice + music files, trigger refs), storyTriggers helper coverage. | ✓MEASURED<br>last 2026-05-03 | 53K<br>tokens / use           | 22K<br>tokens / use           | 15K<br>tokens / use            | 30K<br>tokens / use            |
| <b>equipment</b><br>Create, modify, or remove a weapon or piece of equipment (augment / reactor / shield / armor) â€” including visual changes to how it appears in combat or in the loadout UI.                                                                                                                      | ✓MEASURED<br>last 2026-05-03 | 401K<br>tokens / use          | -5K<br>tokens / use           | 4K<br>tokens / use             | 9K<br>tokens / use             |

|                                                                                                                                                                             |                              |                      |                     |                     |                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------|---------------------|---------------------|---------------------|
| <b>modular-architecture-audit</b><br>Orchestrates the multi-phase modular-architecture audit + refactor.                                                                    | ✓MEASURED<br>last 2026-05-22 | 129K<br>tokens / use | 46K<br>tokens / use | 5K<br>tokens / use  | 10K<br>tokens / use |
| <b>new-enemy</b><br>Scaffold a new enemy â€” enemies.json entry, BootScene placeholder sprite generator, optional test wave, and integrity test verification.               | ✓MEASURED<br>last 2026-05-22 | 68K<br>tokens / use  | 7K<br>tokens / use  | 6K<br>tokens / use  | 11K<br>tokens / use |
| <b>new-migration</b><br>Add a Postgres schema migration end-to-end â€” dated SQL file in db/migrations/, Database interface update in src/lib/db.ts, prod applica...        | ✓MEASURED<br>last 2026-05-22 | 21K<br>tokens / use  | 4K<br>tokens / use  | 7K<br>tokens / use  | 14K<br>tokens / use |
| <b>new-mission</b><br>Scaffold a new combat mission across missions.json, waves.json, the galaxy planet binding, and a smoke test â€” in one shot.                          | ✓MEASURED<br>last 2026-05-22 | 50K<br>tokens / use  | 29K<br>tokens / use | 8K<br>tokens / use  | 16K<br>tokens / use |
| <b>new-perk</b><br>Scaffold a new mission-only perk â€” perks.ts entry, BootScene icon generator, HUD chip wiring, and (for actives) a switch case in PerkCo...             | ✓MEASURED<br>last 2026-05-22 | 56K<br>tokens / use  | 27K<br>tokens / use | 9K<br>tokens / use  | 18K<br>tokens / use |
| <b>new-solar-system</b><br>Add a new solar system to the galaxy â€” solarSystems.json entry (incl. galaxy bed), SolarSystemId union extension, optional unlock gatin...     | ✓MEASURED<br>last 2026-05-22 | 60K<br>tokens / use  | 6K<br>tokens / use  | 13K<br>tokens / use | 26K<br>tokens / use |
| <b>new-story</b><br>Add, modify, or remove story content â€” cinematic popups, voiceovers, music beds, body text, written synopses, and auto-trigger wiring.                | ✓MEASURED<br>last 2026-04-30 | 56K<br>tokens / use  | 14K<br>tokens / use | 5K<br>tokens / use  | 11K<br>tokens / use |
| <b>new-weapon</b><br><i>Superseded by /equipment.</i>                                                                                                                       | REDIRECT                     | —                    | —                   | —                   | —                   |
| <b>save-roundtrip-audit</b><br>Pre-commit save-pipeline invariants check â€” walks every StateSnapshot field through the 8 layers of the save round-trip (snapshot inter... | ✓MEASURED<br>last 2026-05-03 | 19K<br>tokens / use  | 7K<br>tokens / use  | 12K<br>tokens / use | 24K<br>tokens / use |

| SKILL                                                                                                                                                                                                                                                                          | STATUS                               | COST / USE<br><i>measured</i> | SAVE / USE<br><i>measured</i> | COST / USE<br><i>estimated</i> | SAVE / USE<br><i>estimated</i> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------------------|-------------------------------|--------------------------------|--------------------------------|
| <b>ai-codegen-smell-audit</b><br>Read-only audit for specific failure modes that recur in LLM-generated code â€” defensive guards on impossible cases, generic names in domain code, swallowed errors, single-use helpers, mirror tests, and so on.                            | <div>✓MEASURED</div> last 2026-05-17 | <b>199K</b><br>tokens / use   | <b>76K</b><br>tokens / use    | <b>75K</b><br>tokens / use     | <b>150K</b><br>tokens / use    |
| <b>audit</b><br>Run a multi-phase robustness audit of the current codebase.                                                                                                                                                                                                    | <div>✓MEASURED</div> last 2026-05-20 | <b>553K</b><br>tokens / use   | <b>3K</b><br>tokens / use     | <b>425K</b><br>tokens / use    | <b>850K</b><br>tokens / use    |
| <b>mikko-audit-suite</b><br>Run every `mikko-*` audit skill that's relevant to the current codebase, in a sensible order, and produce one index document linking to each audit's report.                                                                                       | <div>✓MEASURED</div> last 2026-05-23 | <b>24K</b><br>tokens / use    | <b>−4K</b><br>tokens / use    | —                              | —                              |
| <b>mikko-help</b><br>List every installed `mikko-*` skill with its description (or with its `barney:` field when `--barney` is passed).                                                                                                                                        | <div>✓MEASURED</div> last 2026-05-20 | <b>644K</b><br>tokens / use   | <b>−5K</b><br>tokens / use    | <b>7K</b><br>tokens / use      | <b>14K</b><br>tokens / use     |
| <b>mikko-install</b><br>Install, update, list, or uninstall `mikko-*` skills from the cloned `claude-skills` source repo into the user-wide skill directory (`~/...`                                                                                                           | <div>✓MEASURED</div> last 2026-05-21 | <b>35K</b><br>tokens / use    | <b>−6K</b><br>tokens / use    | <b>4K</b><br>tokens / use      | <b>8K</b><br>tokens / use      |
| <b>react-anti-patterns-audit</b><br>Read-only audit for six concrete React anti-patterns that recur in component code (key=index in lists, useEffect for derived state, dep-array lies, state mutation instead of replace, missing effect cleanup, multiple sources of truth). | <div>✓MEASURED</div> last 2026-05-23 | <b>22K</b><br>tokens / use    | <b>−5K</b><br>tokens / use    | <b>23K</b><br>tokens / use     | <b>45K</b><br>tokens / use     |
| <b>readme-drift-sync</b><br>Detect drift between what the repo actually contains and what `README.md` claims, then rewrite ONLY the drifted sections in the README's ...                                                                                                       | <div>✓MEASURED</div> last 2026-05-23 | <b>63K</b><br>tokens / use    | <b>−33K</b><br>tokens / use   | <b>40K</b><br>tokens / use     | <b>80K</b><br>tokens / use     |
| <b>security-audit</b><br>Orchestrates the multi-phase security audit + remediation.                                                                                                                                                                                            | <div>✓MEASURED</div> last 2026-05-22 | <b>112K</b><br>tokens / use   | <b>−4K</b><br>tokens / use    | —                              | —                              |

|                                                                                                                                 |                                      |                             |                             |                             |                             |
|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| <b>session-cost</b><br>Measure the token cost of a single Claude Code session — main thread plus every sub-agent it dispatched. | <div>✓MEASURED</div> last 2026-05-21 | <b>409K</b><br>tokens / use | <b>5K</b><br>tokens / use   | <b>8K</b><br>tokens / use   | <b>16K</b><br>tokens / use  |
| <b>skill-calibration</b><br>A/B-test a Claude Code skill against an unstructured baseline to measure actual token savings.      | <div>✓MEASURED</div> last 2026-05-23 | <b>27K</b><br>tokens / use  | <b>-12K</b><br>tokens / use | <b>140K</b><br>tokens / use | <b>280K</b><br>tokens / use |
| <b>skill-usage</b><br>Measure actual Claude Code skill usage from local transcript JSONL files.                                 | <div>✓MEASURED</div> last 2026-05-21 | <b>25K</b><br>tokens / use  | <b>26K</b><br>tokens / use  | <b>4K</b><br>tokens / use   | <b>8K</b><br>tokens / use   |

| <div> <b>mikkonumminen.dev</b> <span>4 skills · 4 measured · <a href="https://github.com/MikkoNumminen/mikkonumminen.dev">github.com/MikkoNumminen/mikkonumminen.dev</a></span> </div>                                                                                |                                      |                               |                               |                                |                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------------------|-------------------------------|--------------------------------|--------------------------------|
| SKILL                                                                                                                                                                                                                                                                 | STATUS                               | COST / USE<br><i>measured</i> | SAVE / USE<br><i>measured</i> | COST / USE<br><i>estimated</i> | SAVE / USE<br><i>estimated</i> |
| <b>md-to-pdf</b><br>Render a markdown report (or any HTML) to a styled PDF using the local Chrome's --print-to-pdf flag.                                                                                                                                              | <div>✓MEASURED</div> last 2026-05-23 | <b>21K</b><br>tokens / use    | <b>15K</b><br>tokens / use    | <b>25K</b><br>tokens / use     | <b>50K</b><br>tokens / use     |
| <b>skill-localUpdate</b><br>Refresh every local artifact the site renders about your portfolio skills — the JSON the /contact terminal fetches at runtime, the measurement-overlaid annual totals, and the downloadable skills-registry PDF — in one sequenced chain. | <div>✓MEASURED</div> last 2026-05-23 | <b>21K</b><br>tokens / use    | <b>23K</b><br>tokens / use    | <b>83K</b><br>tokens / use     | <b>165K</b><br>tokens / use    |
| <b>skill-registry</b><br>Scan every sibling repo under D:/koodaamista for `.claude/skills/*/SKILL.md` files and emit a consolidated registry as a structured JSON ...                                                                                                 | <div>✓MEASURED</div> last 2026-05-20 | <b>148K</b><br>tokens / use   | <b>-6K</b><br>tokens / use    | <b>130K</b><br>tokens / use    | <b>260K</b><br>tokens / use    |
| <b>sync-readmes</b><br>Audit project data against sibling repos' READMEs and open a PR with drift corrections.                                                                                                                                                        | <div>✓MEASURED</div> last 2026-05-18 | <b>119K</b><br>tokens / use   | <b>-4K</b><br>tokens / use    | <b>141K</b><br>tokens / use    | <b>282K</b><br>tokens / use    |

# Method, in plain language

How the numbers in this document are produced, what they mean, and where I am still guessing.

## Per single use — the only unit on this page

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Every number in the document is per one invocation of the skill. There are no annual figures, no monthly figures, no "tokens per year" anywhere. The original document had those; they were models stacked on guesses about how often a skill gets used. This version drops them. If you want to know what a skill costs you over a year, multiply the per-use number by however many times you'll actually run it — the model wasn't going to be more honest than that anyway.

## How “measured cost / use” is produced

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Two sources, both real. **Transcript measurement:** every Claude Code session writes a JSON-Lines transcript to `~/.claude/projects/<dir>/<sessionId>.jsonl`, with each assistant message carrying an `attributionSkill` field when a skill is active. The `skill-usage` skill walks those files, groups by skill, sums `input_tokens + output_tokens + cache_creation_input_tokens` (cache reads excluded — those are paid upstream), dedupes by `requestId`, and reports per-use averages across whatever invocations landed in the 90-day window. That's "what happened in production."

**A/B calibration arm-B:** when a skill hasn't been invoked in production (no transcript data), an A/B run still spends real tokens — a Sonnet sub-agent followed the `SKILL.md` end-to-end on a representative task, with usage billed through the harness. Arm-B IS a real cost-per-use measurement, just from a controlled run instead of in-the-wild use. Rows in this state show the cost with a "tokens / use (A/B-measured)" sub-label. Rows with both transcript and A/B data prefer the transcript number — it's what happened, not what's reproducible.

## How “measured save / use” is produced

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One source: a calibration A/B test. Two Sonnet sub-agents solve the same task in fresh sandboxed worktrees — arm A cold (no `SKILL.md` access), arm B following the skill. Save / use is arm-A tokens minus arm-B tokens for that one run. **N = 1 per skill, single data point.** A re-run would produce different absolute numbers for both arms; trust direction and rough magnitude, not two-significant-digit precision.



Some skills show negative save / use in orange. Those are real findings: the skill arm spent MORE tokens than the unstructured arm, because the skill encodes rigor (e.g. a full-CRUD lifecycle or a multi-phase audit) that the unstructured arm skipped. The skill's value is completeness, not token compression. The arm-A / arm-B numbers are preserved on each calibrated row's receipt for any downstream consumer that wants to see both sides.

## How “estimated cost / use” is produced

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I made up the number. I imagined what running the skill should cost in tokens and wrote that down. There's no math behind these. They are guesses by the person who built the skill, written down before any measurement existed.

The estimates stay in the document so the picture covers every skill, not just the ones I've measured. Every measured estimate I had has turned out to be off, usually low by 5× to 100× — apply the same skepticism to the rows that haven't been measured yet. If anything, the un-measured estimates are likely to be *more* wrong: I measured the ones I felt most confident about first.

## How “estimated save / use” is produced

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3× heuristic on the estimated cost: *saved*  $\approx$  2× *cost*. The model says "doing this without the skill would cost about 3× what the skill costs," so the saved-per-use is 2× the cost-per-use. The 3× is a handful-of-side-by-side-runs guess; the May-2026 Spacepotatis calibration showed it's overstated by roughly 3× at the portfolio level (measured savings averaged ~22% of arm-A cost, vs the heuristic's ~67%). Treat estimated savings as a possibly-too-optimistic upper bound.

Cost is on both sides of the underlying subtraction, so the estimated save figure is more sensitive to bad cost estimates than the estimated cost is. An italic estimated-save stacked on top of an italic estimated-cost is a model on top of a guess — least trustworthy cell on the page. A green measured-save on top of a green measured-cost is the most trustworthy. The visual treatment matches that hierarchy.

## What this document does NOT claim

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- No production cost numbers. This is development-time tooling on my own machine.
- No comparison to other engineers' setups. I only have transcripts for one engineer.
- No per-feature ROI claim. This is per-skill cost, not per-feature value.

- No assertion that any of this scales linearly to a team. It might. I have not measured.
- No claim that the modeled savings would survive an actual A/B test against well-prompted unstructured chats. They might; I haven't tested.

## Reproducibility

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From inside this repo ( `mikkonumminen.dev/` ) on a machine with Claude Code installed:

1. `/mikko-skill-usage` — writes `.claude/agent-verdicts/SKILL-USAGE-LATEST.json` from local transcripts.
2. `/skill-localUpdate` (formerly `/skill-pdf` ) — re-walks the portfolio inventory, applies the measurement overlay, renders this PDF.

The data the renderer reads is committed to the repo at `public/data/skills-registry.json` . If you want to inspect what's behind a number on this page, that file is the source of truth. The dated history lives in `.claude/agent-verdicts/SKILL-REGISTRY-*.json` .

The *last* `<date>` marker on a measured row is the timestamp of the most recent Claude Code transcript that invoked that skill. The transcript itself stays on my machine because it contains code and context from private repos. If you want to verify the measurement methodology rather than the measurement, the parser source is at `~/.claude/skills/mikko-skill-usage/SKILL.md` — it is a JSONL parser, not a network call to anything.